

Shower Device

Happy Campers

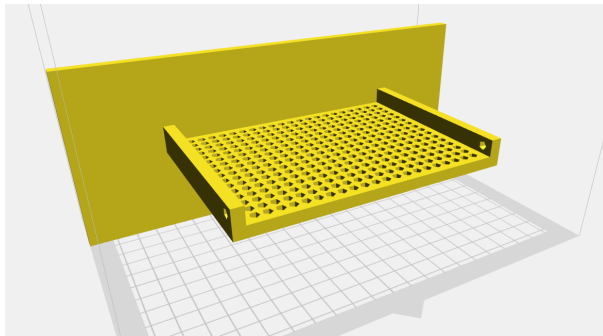
Element G: Construction of a Testable Prototype

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

The first prototype was made out of cardboards and duct tape, the team measured and cut the cardboard and had them taped together and have a 1:1 model with only difference in the thickness of back plate and base plate, which differs from the designed 1 inch to one cardboard's thickness. Therefore the team could test the area capacity of the prototype without wasting too much time and cardboard. One change the team made is to abandon the one piece though hole in the wall plan since it is not permitted to drill a hole in the middle of the shower divider, and used a backup plan that does not require drilling.

The second prototype was a 3D printed 1:1 model, using PLA filament and having nylon rope attached. The only difference between the final prototype is it is using

only 15% infill while printing in order to shorten the print time and save filament. The team created a 3D model of the prototype, and sliced it using the dedicated slicing software for the printer, and had it printed. One significant change the team made is changing the model to four separate pieces and having them connected after being printed out instead of printing the prototype as a whole. Additionally, the team resized the prototype to 95% of its original design. Both changes are a compromise to the rather small printing platform of the 3D printer (Flashforge Creator 3 Pro).



The expert the team found, Mr. S, is the physics teacher at our school. Since the team is mainly testing the weight capacity and balance of the prototype, he is absolutely qualified to inspect the team's design draft and analyse it as a free body diagram. According to Mr. S, the design was without any significant mechanical errors, although the middle where two 3D printed pieces connect will be a potential failure. Additionally, he pointed out the connection will be easy to get moulded because of the existence of a small gap.

The team could not test the prototype's weight capacity as the final design since it is using a significantly smaller infill percentage.